

FINANCIAL RISK & FRAUD DETECTION

FINGUARD

AI



A hybrid rules, machine-learning and behavioral-analysis platform for scoring financial transactions in real time — and explaining every decision it makes.

DETECT · EXPLAIN · INVESTIGATE · PREVENT

THE CORE PROBLEM

RULE ENGINES ALONE
CAN'T KEEP UP

Legacy systems flag transactions against a fixed rulebook. That approach breaks down in predictable ways: it can't see patterns it hasn't been told to look for, it weighs every signal the same, and it can't say why it blocked a payment.

OBJECTIVES

- Real-time risk scoring
- Hybrid fraud detection
- Explainable decisions
- Automated investigation
- Continuous model improvement

"Detection is one subsystem, not the whole platform. We needed a pipeline that explains itself."

FAILURES OF STATIC RULES

- 01

Static rules

New fraud patterns go undetected until someone manually writes a rule for them.
- 02

High false positives

Blunt thresholds block legitimate customers as often as they catch bad actors.
- 03

Delayed detection

Batch-style checks catch fraud after the money has already moved.
- 04

Manual investigation

Every flagged case still needs an analyst to piece together the evidence by hand.
- 05

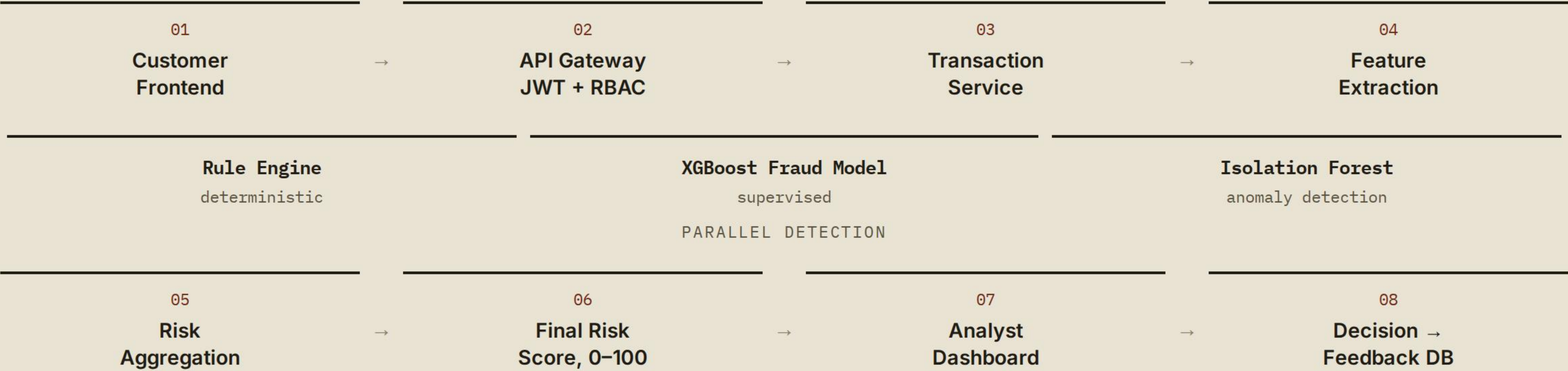
Poor adaptability

Limited contextual analysis and little explainability behind each decision.

THE FINGUARD APPROACH

FinGuard layers deterministic rules, a supervised fraud model, unsupervised anomaly detection, behavioral analysis and explainability into a single real-time scoring pipeline.

TRANSACTION TO DECISION, ONE CONTINUOUS PIPELINE



RISK FORMULA WEIGHTING

KEY ML FEATURES

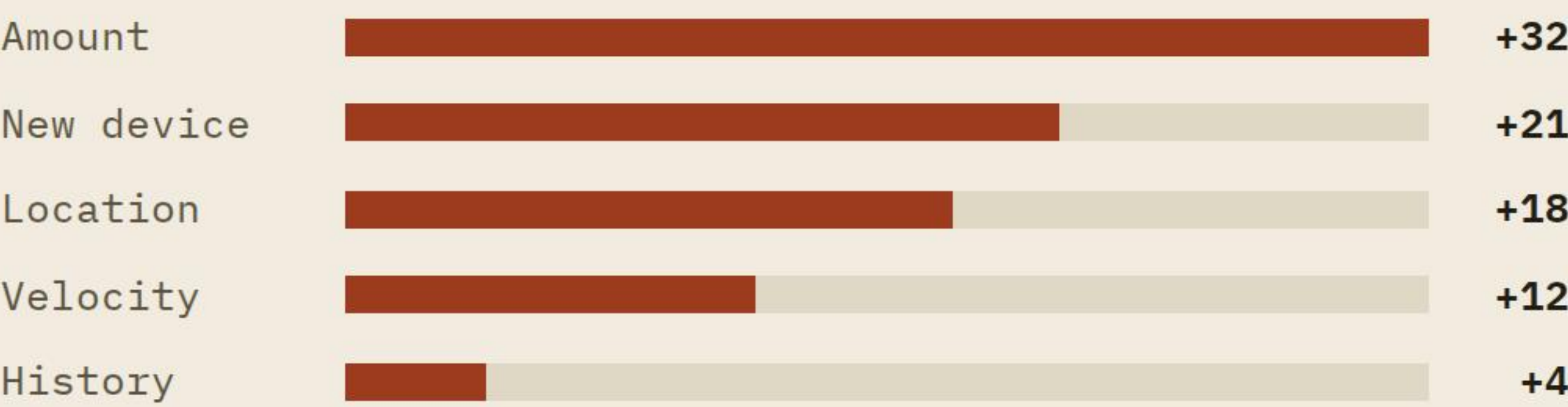
- Amount
- Velocity
- Geography
- Device

EXPLAINABLE AI – WORKED EXAMPLE

EXHIBIT – TRANSACTION REVIEW

Amount ₹85,000 • normal range ₹5,000 – ₹10,000

SHAP contribution to score



Fraud probability **91%**
Risk score **87 / 100**



BUSINESS CASE: BEFORE & AFTER

BEFORE FINGUARD

Static rules only

Manual investigation

Delayed detection

High false positives

WITH FINGUARD AI

Real-time risk scoring

Hybrid ML + rules

Explainable decisions

Continuous learning

VALUE DELIVERED (KPIs)

Fraud Recall

False Positive Rate

Mean Time to Detect

Analyst Productivity

Fraud Loss Avoided

FINGUARD TURNS FRAUD DETECTION FROM A STATIC, RULE-BOUND
PROCESS INTO AN ADAPTIVE, EXPLAINABLE RISK PLATFORM.